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Optimized Feature Selection and SHAP-Based Explainable ML for Celiac Disease Prediction

Md Mahbubur Rahman Druvo
Department of Computer Science and Engineering
Dhaka International University
Dhaka, Bangladesh
drubo668@gmail.com

Abir Chowdhury
Department of Computer Science and Engineering
Dhaka International University
Dhaka, Bangladesh
abir.chowdhury.sami1@gmail.com

Md. Shariful Islam
Department of Computer Science and Engineering
Dhaka International University
Dhaka, Bangladesh
shariful.cse@diu.ac

Md. Tahzib Ul Islam
Department of Computer Science and Engineering
Dhaka International University
Dhaka, Bangladesh
tahzib.cse@diu.ac

Md. Abdul Based
Department of Computer Science and Engineering
Dhaka International University
Dhaka, Bangladesh
based@diu.ac

Khandaker Mohammad Mohi Uddin*
Department of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
jilanicsejnu@gmail.com

***Abstract*— Celiac disease is a chronic autoimmune illness caused by gluten ingestion that requires an early and correct diagnosis to be treated effectively. Traditional diagnostic procedures are sometimes imprecise, necessitating machine learning-based solutions. This study proposes a comprehensive methodology for celiac disease prediction based on clinical and laboratory data. To maintain data consistency, the preprocessing pipeline scales features and encodes categorical variables. The model's performance was optimized by applying chi-square feature selection to find the most pertinent attributes. Using an 80:20 data split, five machine learning models were trained and evaluated. The model's performance was measured using F1-score, AUC score, accuracy, precision and recall. KNN was the most successful in predicting celiac disease, with the highest accuracy of 99.23%, AUC score of 98.07% and precision of 99.04% among the tested models. The interpretability of the model was further assessed using SHAP analysis, shed light on the significance of the features. This work shows machine learning can potentially use to improve the detection of celiac disease, providing a reliable and scalable solution for medical applications.**

Keywords— Celiac disease, chi-square feature selection, machine learning, interpretability, SHAP

I. Introduction

A chronic autoimmune condition that affects the small intestine in people who are genetically predisposed is called celiac disease (CD) [1]. The consumption of gluten, a protein present in wheat, barley and rye, causes it. The immune system of a person with celiac disease reacts to gluten consumption by attacking the small intestine, causing inflammation and harm to the villi, which are microscopic projections that resemble fingers and are in charge of absorbing nutrients [2]. The loss of vital nutrients brought on by this damage can result in a number of symptoms, such as persistent diarrhea, bloating, stomach pain, exhaustion, anemia and even neurological problems including melancholy and migraines. Because of its many and occasionally vague symptoms which can mimic those of other gastrointestinal conditions like inflammatory bowel disease (IBD) and irritable bowel syndrome (IBS), celiac disease is frequently underdiagnosed [3]. Although intestine biopsies and blood tests are still the most common diagnostic techniques, they can be time-consuming and may not always yield a conclusive diagnosis in instances that are borderline [4]. Serious long-term consequences like osteoporosis, infertility, neurological issues and an elevated chance of intestinal malignancies can also result from a delayed diagnosis. Advances in machine learning (ML) and artificial intelligence (AI) hold promise for bettering celiac disease diagnosis and early detection [5]. Large datasets can be analyzed by machine learning models which can also improve diagnostic precision and spot intricate patterns. ML-based methods can help medical personnel make faster and more accurate diagnoses by combining clinical data, biomarker analysis and feature selection algorithms. In an effort to close the gap between conventional diagnostic techniques and data-driven insights, this study investigates the use of machine learning algorithms to enhance celiac disease detection and classification.

Make a point below on the contribution of our effort.

- Benchmark five popular classifiers using a uniform preprocessing and feature-selection methodology, offering a clear baseline framework for upcoming machine learning research on celiac disease.
- The most useful pre-diagnostic clinical and serological predictors are found using a Chi-square-based ranking method that increases efficiency without sacrificing interpretability.
- SHAP was integrated to provide insight into how individual factors affect model predictions.
- Provided a scalable solution to reduce the diagnostic gap, improving patient outcomes and healthcare effectiveness. To avoid label leakage and ensure a clinically realistic and repeatable prediction setting, implement a strict feature-use methodology that eliminates post-diagnosis or outcome-dependent factors.

II. Related Work

Hedesh et al. [6] used a dataset of 50 suspected cases (70% positive, 30% negative) to apply machine learning for early

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celiac disease diagnosis. Ten-fold cross-validation was used to train five models. 88% recall and 80% accuracy were attained by a metaclassifier that used majority voting. Age, lactose intolerance, persistent diarrhea, gluten sensitivity and stomach pain were important predictive factors that demonstrated how machine learning may increase the precision of diagnosis. Wichka et al. [7] used machine learning to design drugs that target the suppression of transglutaminase 2 (TG2) in celiac disease. The accuracy of boosting ensembles of deep learning and random forest models was above 90% using data from 1100 assays. A regression model was used to rank the active substances and identify key chemical properties. Celiac Informatics is a web tool designed to help with therapy research by evaluating drug-likeness and molecular screening. Jinnuri et al. [8] compared ML models to standard celiac disease diagnostics to improve early identification while lowering invasiveness. It uses random forest, gradient boosting, SVC and logistic regression to discover essential features such as age, diabetes and abdominal pain. Effective feature selection enhances model dependability, enabling earlier nutritional interventions and better patient outcomes. Maya Silvi Lydia et al. [9] created a celiac disease diagnosis method used Kaggle numerical data and an artificial neural network model (ANN). The model was tuned using adam and categorical cross-entropy, achieved 97.07% accuracy. A web-based tool was developed to facilitate diagnosis. Francesco Piccialli et al. [10] used machine learning (ML) to forecast when potential celiac disease (PCD) will develop into overt celiac disease (CD). The accuracy of ML models using a dataset of PCD patients was over 75% with optimal boosted trees achieving the highest accuracy of 80%. The models reduced the number of features from 85 to 19. Because it identifies high-risk patients for early intervention, this method improves precision medicine. Michael Dreyfuss et al. [11] used anonymized outpatient data, researchers created machine learning models to identify individuals at risk for undiagnosed celiac disease (CD). Five models were tested with the highest performing being XGBoost (AUC = 0.86), followed by logistic regression (AUC = 0.85) and random forest (AUC = 0.83). Sana Syed et al. [12] a CNN-based study demonstrated the potential of AI in enhancing CD diagnosis by achieving 93.4% accuracy in differentiating between healthy tissues, environmental enteropathy (EE) and celiac disease (CD) from duodenal biopsies. Agrima Mehandiratta et al. [13] used machine learning to predict CD and investigates its relationship with Type 1 and 2 diabetes and thyroid Disease. Its goal is to improve patient outcomes by harnessing medical databases for early identification. Alina Delia Călin [14] used clinical signs and biomarkers, this study uses artificial intelligence to predict the diagnosis of CD. Machine learning algorithms classified six disease subtypes with 92% accuracy and distinguished CD cases with 97% accuracy. The results demonstrate AI's potential in CD diagnosis despite research constraints. Data sharing among researchers is encouraged to improve the robustness and accuracy of the models.

III. Methodology

In Fig. 1 the dataset collected from Kaggle [15]. It was preprocessed to ensure consistency and machine learning method compatibility by scaling numerical features and encoding categorical variables. The Chi-square test was used for feature selection in order to find the most pertinent predictors, lowering dimensionality and concentrating on significant factors. Dataset divided by 80:20 ratio. In proposed system, trained and assessed five machine learning models: k-nearest neighbors (KNN), bernoulli naive bayes (BNB), multinomial naive bayes (MNB), quadratic discriminant analysis (QDA) and gaussian naive bayes (GNB). Metrics like precision, recall, F1-score, accuracy and AUC-score were used to evaluate model performance and identify the top performing algorithm. Model predictions were interpreted using SHAP values which also offered transparency by highlighting each feature's contribution. After analyzing the data to uncover important predictors and offer insights into the dataset, recommendations were made in because of the results.

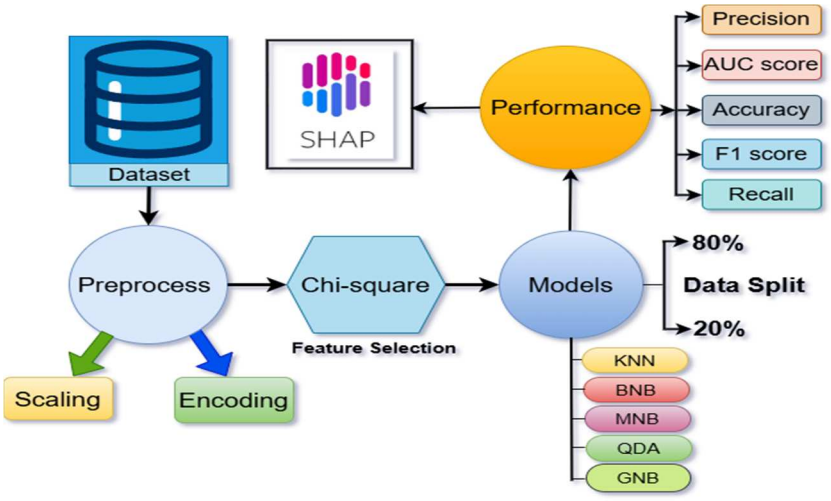


Fig. 1. Workflow of the proposed work

A. Dataset

The dataset has 2,206 records and 15 characteristics, includes clinical and laboratory data on celiac disease such as demographic information, symptoms, diagnostic markers and disease outcomes. Patient age, gender, diabetes presence and type, diarrhea, gastrointestinal pain, short stature c
sticky stool, weight loss and immunoglobulin leve
IgM) are all important factors. It also contains the Marsh classification for intestinal damage severity and celiac disease form and the ultimate disease diagnosis (Yes or No). Only pre-diagnostic clinical and serological characteristics were employed as predictors to guarantee a fair and clinically realistic prediction setup.

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B. Data processing

The preprocessing stage, the dataset went through some key stages to ensure its appropriateness for analysis [16]. Labels were used to encode categorical information including gender, diarrhea type and celiac disease categorization into numerical representations. Standardization was used to scale numerical features such as age and immunoglobulin levels (IgA, IgG and IgM) in order to ensure consistency and enhance the functionality of distance-based algorithms. Bernoulli and Multinomial Naïve Bayes models were kept for comparison benchmarking in order to assess their empirical behavior on continuous clinical data, despite their assumption of binary or count-based feature distributions. A more theoretically aligned reference is offered by the Gaussian Naïve Bayes (GNB) model, which naturally handles continuous information.

C. Feature Selection

The most pertinent variables for diagnosing celiac disease were found through feature selection, ensuring the effectiveness and interpretability of the model. Seven features were chosen based on their influence on the target variable using the chi-square approach. 'Diabetes' designates the presence of diabetes, 'Diabetes Type' designates the type of diabetes, 'Abdominal' indicates abdominal pain, 'Weight_loss' indicates weight loss, 'IgA' captures immunoglobulin A levels, 'Marsh' describes the amount of intestinal damage according to the Marsh classification and 'cd_type' describes the type of celiac disease. Before using the chi-square feature selection test, continuous numerical characteristics (Age, IgA, IgG, and IgM) were discretized into quartile-based bins to guarantee appropriate statistical validity. The test's need for non-negative, categorical-style data is ensured by this preprocessing step. These characteristics were selected due to their high association with the target variable of relevance. Seven features selected based on chi-square score and P-value in TABLE I.

TABLE I. FEATURE SELECTION BY CHI-SQUARE SCORE.

Feature	Chi-squared Score	P-value	Selected
Marsh	376.7198	6.44E-84	TRUE
Diabetes Type	323.6065	2.37E-72	TRUE
Diabetes	229.1112	9.31E-52	TRUE
cd_type	11.1919	0.000822	TRUE
Weight_loss	6.406874	0.011368	TRUE
IgA	3.476936	0.06223	TRUE
Abdominal	2.1447	0.143063	TRUE
Age	2.091828	0.148089	FALSE
Sticky_Stool	1.476508	0.224322	FALSE
Short_Stature	0.782908	0.376253	FALSE
Gender	0.276265	0.59916	FALSE
IgG	0.233836	0.628694	FALSE
Diarrhoea	0.07171	0.788863	FALSE
IgM	0.043992	0.833869	FALSE

D. Data Split

The dataset is separated into training and testing subsets to effectively evaluate model performance with 80% used for training and 20% for testing [17]. This ensures that models learn on a part of the data while being tested on previously unseen data to assess generalization and minimize overfitting. The train_test_split function from sklearn is used for this, with features (X) and the target variable (y) separated and stratified sampling to ensure consistency in the target class distribution across subsets.

E. K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a non-parametric, distance-based technique for classification and regression. It groups data points according to the majority class of their closest neighbors, which is established by distance measures such as Euclidean [18]. The number of neighbors (k), distance metric, and weighting scheme hyperparameters were optimized for the K-Nearest Neighbors (KNN) classifier using a grid search with five-fold cross-validation. With k = 5, metric = Minkowski, and weights = distance, the optimal configuration was found. To avoid data leaking, feature scaling and imputation were carried out within a single pipeline. It works following the formula (1).

$$\text{Distance Between A and B} = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2} \quad (1)$$

IV. RESULT AND DISCUSSION

Recall assesses the capacity to identify real positives, accuracy looks at overall correctness and precision shows the accuracy of positive predictions. The F1 Score evaluates classification performance across a wide range of thresholds, while ROC-AUC balances precision and recall [19]. The framework uses high-performance CPUs to train complex machine learning models and handle large datasets. True positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) are the words used to define the following numbers, we were able to calculate the performance parameter: (2), (3), (4), (5), and (6).

$$Accuracy = \frac{TN + TP}{TP + TN + FP + FN} \quad (2)$$

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

$$F_1 - Score = 2 * \left(\frac{Recall * Precision}{Recall + Precision}\right) \quad (5)$$

$$AUC = \sum_{i=1}^{n-1} (FPR_{i+1} - FPR_i) * \frac{(TPR_{i+1} - TPR_i)}{2} \quad (6)$$

A. Confusion Metrics

In Fig. 2 the confusion matrices show well the models perform for selected features. The most effective of them is KNN (b), which achieves 104 true positives and 0 false negatives, indicating extremely accurate classification. Despite its good performance, QDA (a) has two false positives and five false negatives which somewhat lowers its reliability. With MNB misclassifying 11 cases as false negatives and 9 as false positives, and BNB(d) likewise having trouble with 9 false positives and 9 false negatives, MNB (c) and BNB (d) exhibit poorer accuracy. KNN performs better than the other models overall, making it the best model for the classification task at hand.

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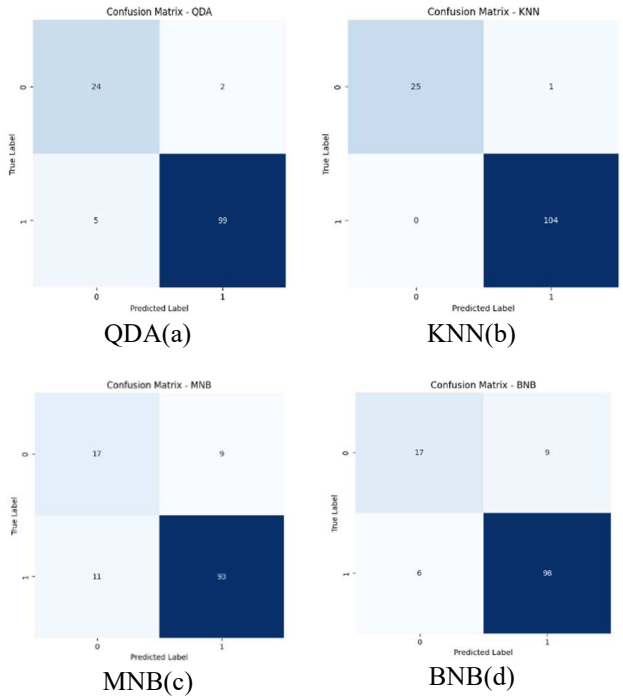


Fig. 2. Confusion metrices with selected features.

In Fig. 3 the confusion matrices show classification accuracy by comparing models for all features. KNN (a) ensures high accuracy and no missed positive cases by performing best with 98 TP, 27 TN, 5 FP and 0 FN. With 98 TP, 26 TN, 6 FP, and 0 FN, QDA (b) comes in second but marginally lower because of a greater false positive rate. Both MNB (c) and BNB (d) exhibit poorer performance with increased misclassification at 92 TP, 22 TN, 10 FP and 6 FN for MNB and 93 TP, 21 TN, 11 FP and 5 FN for BNB. Naïve bayes models perform poorly overall, while KNN and QDA are the most effective models.

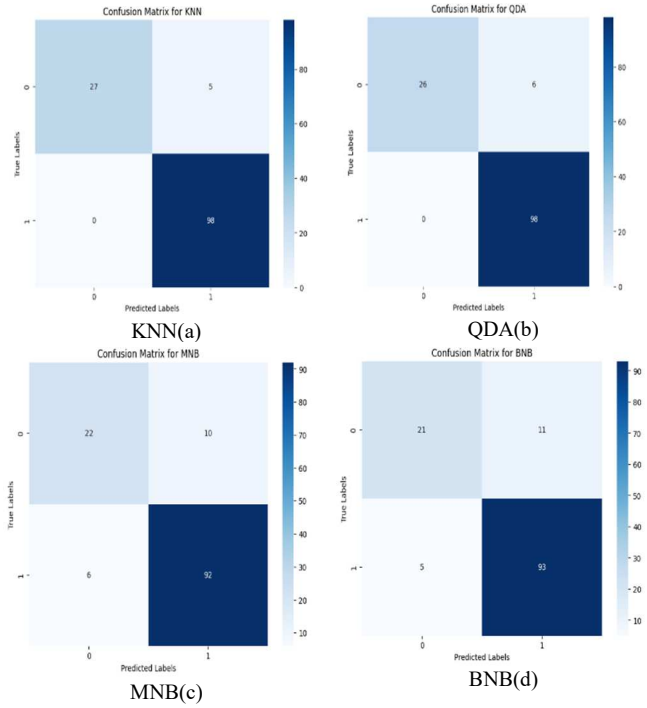


Fig. 3. Confusion metrices with all features.

B. Roc Curve

In Fig. 4 the ROC curve evaluates various machine learning models perform in terms of classification efficacy. K-nearest neighbors (KNN) has the best classification skill with the greatest AUC of 1.00. Following closely AUC of 0.96, Gaussian Naïve Bayes (GNB) exhibits significant predictive capability. Additionally, quadratic discriminant analysis (QDA) exhibits a strong classification performance with an AUC of 0.93. The AUC values of 0.87 for bernoulli naïve bayes (BNB) and multinomial naive bayes (MNB) are lower, indicating poorer performance in comparison to the other models. In this comparison, KNN is the most successful model, whilst MNB and BNB have the least discriminative power.

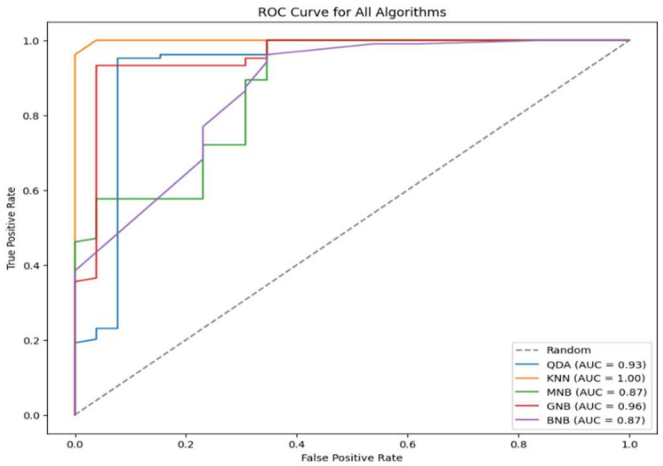


Fig. 4. Roc Curve for all models with selected features.

In Fig. 5 the ROC curve uses the AUC values of various machine learning models with all features to compare perform in categorization. K-nearest neighbors (KNN) has higher classification accuracy with the greatest AUC of 0.97. Following with an AUC of 0.90, gaussian naive bayes (GNB) demonstrates a high capacity for prediction. Quadratic discriminant analysis (QDA) exhibits dependable classification ability despite a little lower AUC of 0.89. The models that perform the worst in this comparison are bernoulli naive bayes (BNB) and multinomial naïve bayes (MNB) which have the lowest AUC values of 0.88 and 0.87, respectively. MNB and BNB have comparatively lower discriminative capability, whereas KNN is the most resilient model.

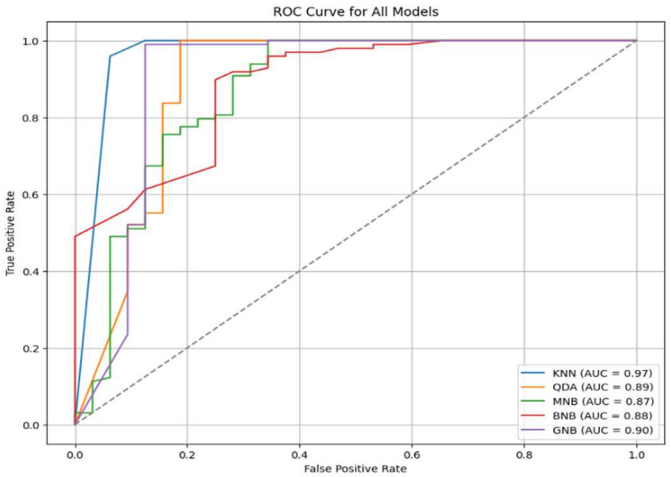


Fig. 5. Roc Curve for all models without selected features.

C. Figures and Tables

In TABLE II, KNN model is the most dependable classifier, with an accuracy of 99.23%, perfect recall (100%) and the greatest AUC score (0.98). As a competing option, quadratic discriminant analysis (QDA) comes in second with 94.61% accuracy, showing excellent performance with high precision (98.02%) and recall (95.19%). Both gaussian naive bayes (GNB) and bernoulli naive bayes (BNB) exhibit balanced with reasonably good recall with respective

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accuracy rates of 90% and 88.46%. Multinomial naive bayes (MNB). On the other hand, performs worse than the other models with the lowest accuracy of 84.61%. All things considered, KNN is the best model, QDA is still a good substitute, and MNB is the worst.

TABLE II. MACHINE LEARNING MODEL PARAMETERS SCORE WITH SELECTED FEATURES

Model	Accuracy	Precision	Recall	F1-score	AUC Score
QDA	0.946154	0.980198	0.951923	0.965854	0.9375
KNN	0.992308	0.990476	1	0.995215	0.980769
MNB	0.846154	0.911765	0.894231	0.902913	0.774038
GNB	0.90	0.925234	0.951923	0.938389	0.822115
BNB	0.884615	0.915888	0.942308	0.92891	0.798077

In TABLE III, KNN continues to be the top-performing model with all features are taken into account with the highest accuracy of any model at 96.15%. It is the most dependable classifier due to its high precision (97.57%) and AUC score (0.97). A competitive option, QDA performs well with precision (97.11%) and AUC score (0.89), following closely behind with 95.38% accuracy. With an accuracy of 91.54%, GNB has a reasonable level of performance, balancing recall and precision. In comparison to the best models, MNB and BNB have the lowest accuracy (87.69%), as well as worse precision and recall scores. Overall, KNN continues to be the most successful model with QDA coming in second and MNB and BNB performing worse in classification.

TABLE III. MACHINE LEARNING MODEL PARAMETERS SCORE WITH ALL FEATURES

Model	Accuracy	Precision	Recall	F1 Score	AUC Score
KNN	0.961538	0.975728	0.921875	0.945189	0.966199
QDA	0.953846	0.971154	0.90625	0.933424	0.888393
MNB	0.876923	0.843838	0.813138	0.826667	0.868782
BNB	0.876923	0.850962	0.802615	0.822465	0.883929
GNB	0.915385	0.949541	0.828125	0.869656	0.900032

In Fig. 6 the accuracy of several machine learning models employing certain features is shown in the bar chart. KNN performs noticeably better than the other models, achieving the greatest accuracy of 99%. Second-best performer QDA comes in second with 95% accuracy. GNB performs moderately, achieving 90% accuracy, while BNB and MNB fall short at 88% and 85%, respectively. This suggests that KNN is the best model for classification, followed by QDA with MNB and BNB performing the worst.

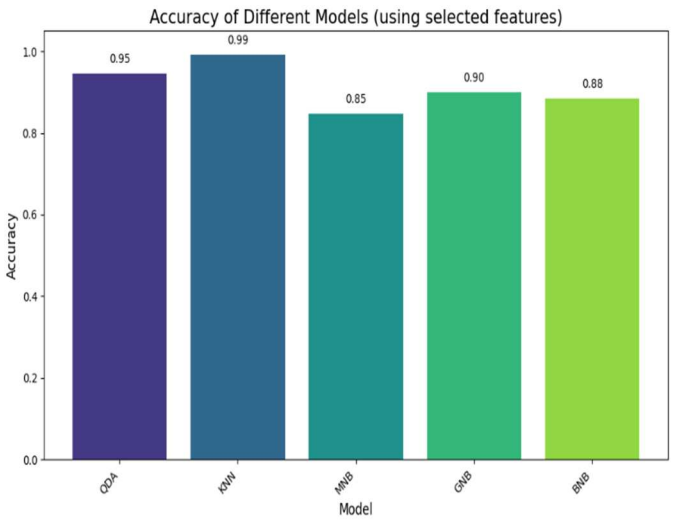


Fig. 6. Five models accuracy score for selected features.

In Fig. 7 five models are compared in terms of accuracy based on all features in the bar chart. KNN attains the highest accuracy of 0.96 with QDA coming in second at 0.95. At 0.92, GNB performs marginally worse than MNB and BNB, which have the lowest accuracy at 0.88. All models perform well, with accuracies above 0.88, however KNN and QDA are the models that perform the best overall, while MNB and BNB have worse accuracy.

In TABLE IV, A comparison of our suggested model with previous studies in the field of celiac disease detection. It is clear that despite the encouraging outcomes of earlier research employing a variety of datasets and machine learning algorithms, our method performs noticeably better in terms of accuracy. Hedesh et al. [6] attained an accuracy of 80.00%, which is much lower given the small dataset size. Similarly, Francesco Piccialli et al. [10] achieved 86.00% using an extremely randomized trees (ERT) model but their dataset was rather small (340 patients). Other studies, like Ameya Jinnuri et al. [8] (95.00%) and Ibrahim Wichka et al. [7] (96.00%), used larger datasets and more complex models such as random forest and adaboost. Maya Silvi Lydia et al. [9] recorded the highest prior accuracy, 97.00%, using an artificial neural network (ANN). However, our proposed work establishes a new standard for accuracy, attaining an astonishing 99.23% utilizing the k-nearest neighbors (KNN) model on a large dataset of 2206 patients sourced from kaggle. This outstanding performance demonstrates the efficacy, resilience and excellence of our automated celiac disease detection method. Our strategy not only improves diagnostic accuracy but also increases reliability, making it a game-changing innovation in the use of machine learning for medical diagnosis.

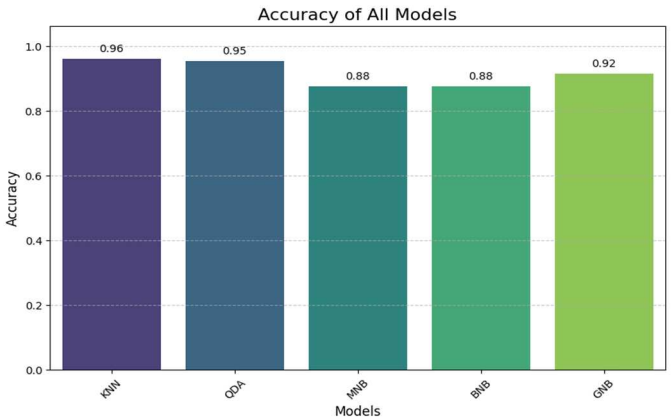


Fig. 7. Five models accuracy score with all features.

TABLE IV. PERFORMANCE ANALYSIS WITH OTHER WORKS

References	Dataset	Model	Accuracy (%)
Mahshid Mohsenzadeh Hedesh et al. [6]	50 patients dataset	×	80.00
Ibrahim Wichka et al. [7]	1130 patients dataset collected from NCBI	AdaBoost	96.00
Ameya Jinnuri et al. [8]	Make a dataset from multiple dataset	Random Forest	95.00
Maya Silvi Lydia et al. [9]	2206 patients dataset collected from kaggle	ANN	97.00
Francesco Piccialli et al. [10]	340 patients dataset	ERT	86.00
Our Proposed	2206 patients dataset collected from kaggle	KNN	99.23

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D. Interpretability

In Fig. 8 the contribution of each feature is represented by its SHAP value, and the SHAP summary graphic how particular features affect the model's predictions. The SHAP value which indicates how much a feature pushes predictions towards good or negative outcomes, is displayed on the x-axis. Predictions are greatly impacted by features like Marsh and cd_type which have the biggest impact and a wide range of SHAP values. Although they have differing degrees of impact IgA, weight loss, diabetes, diabetes type and abdominal also play a role. Red indicates high feature values whereas blue indicates low feature values. By emphasizing the most significant predictors, this analysis aids in the interpretation of the model's decision-making process.

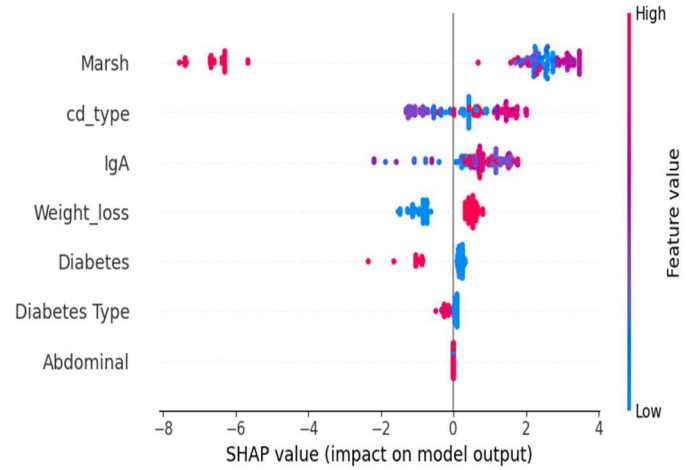


Fig. 8. SHAP analysis key features contributing to model predictions.

V. Conclusion

The limitations of conventional diagnostic techniques are addressed in this study by presenting a machine learning-based for the precise prediction of celiac disease. To provide high-quality data for model training, a comprehensive preprocessing pipeline comprising feature scaling and categorical encoding was put into place. The most essential features were chosen using Chi square which improved the model's performance. With the highest accuracy of 99.23% among the five machine learning models that were assessed, KNN was the most dependable prediction. Furthermore, SHAP analysis was included to improve the interpretability of the model and offer insightful information on the significance of the features. This work opens the door for more effective, scalable and non-invasive diagnostic methods in the medical profession by proving the effectiveness of machine learning in celiac disease prediction.

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